

These are explained as follows:

First, as shown in Equation 10.16 the complete portfolio optimization is only based on a single risk aversion parameter. This ensures consistency across the investment and trading decisions. A consequence of this formulation is that any trading strategy derived using a risk aversion parameter that is different than that used during portfolio construction will result in lower investor utility since it would not correctly quantify trading risk with investment risk. This is most notable for a VWAP strategy where risk aversion is set to be $\lambda = 0$ and results in higher risk and lower utility.

Second, notice that the expression in the trading horizon section of Equation 10.17 is not dependent upon any benchmark price. Indirectly it is based on the current price since we are starting with the current portfolio value. Thus, any post-trade analysis based solely on a specified benchmark price or computed as the difference between average execution price and some benchmark is not the ideal approach to evaluate a trader's performance or skill because it does not consider the underlying goal of the manager or trader. Since the newly formulated portfolio optimization is now based on an expected market impact cost and corresponding trading risk estimates, subsequent post-trade performance attributions need to incorporate these values in order to be able to provide any meaningful benefits. It is, however, essential that post-trade analysis be performed to assess the accuracy of the market impact and trading risk estimates to ensure appropriate future investment decisions. Furthermore, with the advent of algorithmic trading, algorithms based on achieving a specified benchmark price rather than a specified cost will surely hinder overall portfolio performance. For example, a VWAP strategy will likely increase risk exposure and reduce overall utility.

Third, portfolio construction with trading costs needs to be formulated as a multi-period optimization problem. This requires both a trading period that will accommodate a market impact estimate based on size, volatility, and composition of the trade list (e.g., diversified market impact effect), and an investment holding period where there will not be additional changes to the portfolio. As shown above, to achieve the maximum benefit it is essential that the market impact cost be based on both size and strategy. Thus allowing managers to implement their decisions in an appropriate manner—aggressive, passive, or normal—depending upon the risk composition of the trade list.

Fourth, the portfolio manager's and trader's decisions are not separable. A decision making framework that first maximizes the risk-return profile in the investment problem then minimizes trading costs is not guaranteed